

IC Lab Final 2021 Spring

Name:

Student ID: _____

Account: iclab

Total score: 100%

1. [10%]:

Please clearly explain how double flop synchronizer and XOR double flop synchronizer work in our design if we use them in a CDC (clock domain crossing) design and why do we need to use synchronizer in a CDC design.

2. [4%]:

We inserted synchronizers between multi clock domains. However, without further settings, the slack will be negative in synthesis and timing violation will occur in gate level simulation. Express what action must be done to remove the negative slack issue in synthesis (2%) and timing violation issue in gate level simulation (2%).

3. [13%]:

There are three types of simulation flow supported by the Primetime tool for power simulation. Please explain all of them. You need to describe the **name**, **files of information required for simulation**, and **rank of precision** (write "1" for the most precise simulation, write "3" for the least precise simulation). Also, there are two types of signal activity files for power simulation. Please describe them respectively.

Name of the flow (3%)	Required information (3%)	Rank of precision (3%)

Describe the signal activity files (4%):

4. [7%]:

Clock gating can be implemented by either AND-gating or OR-gating. Which one is better?(1%) Why?(3%) Explain why a glitch may happen when we are using clock gating and how to avoid it?(3%)

5. [4%]:

- (1) Please describe the major functionality of "typedef" syntax.
- (2) Please describe the major functionality of "enum" syntax.

6. [10%]:

- (1) Please explain what are pre_randomize() and post_randomize() function in every class? How to use these functions?
- (2) Please write the corresponding code if we want to create a class called "random_val" which contains a random member variable called "rand_val", which is a 32-bits variable. The possible values of this variable are 32'h10000000, 32'h01000000 and 32'h00000001 and these values need to be cyclic randomized without repeating.

7. [12%]:

If we want to check whether our design hits the specification, we can write the assertion in the checker.sv file. There are three signals, A, B, C, in the design, and three signals will only be high consecutive unfixed cycles. Please finish the three assertions below. Notice that you cannot use always block here.

- (1) If A happens, 1 or more cycles later B should be 0 when C happens.
- (2) If A changes from 1 to 0 between one rising clock, then B must happen 10 times in a row after 2-10 cycles.
- (3) If A is 0, B was high before 2 clock cycles.

Assertion1: assertion property (@(posedge clk)) ;

Assertion2: assertion property (@(posedge clk)) ;

Assertion3: assertion property (@(posedge clk)) ;

8. [6%]:

Given the verilog code shown below, calculate the total toggle coverage percentage of register 'data' after simulation. Please briefly explain your reasons.

```
reg [4:0] data;
```

```
initial
```

```
begin
```

```
    data = 8;
```

```
    #50;
```

```
    data = 18;
```

```
    #50;
```

```
    data = 10;
```

```
    #50
```

```
    data = 27;
```

```
    #50
```

```
    data = 19;
```

```
end
```

9. [10%]:

What libraries and files are used in APR. Please explain the purpose of including these files.

10. [6%]:

Explain what's Signal Integrity (SI) Issue, and write at least two solutions.

11. [6%]:

With a more advanced process, does IR drop tend to be more serious or more trivial? List at least two reasons.

12. [12%]:

- a. [4%] What is the difference between power/rail analysis?
- b. [4%] What input files do we need when running vector-based power analysis and rail analysis respectively? What's the content in the files?
- c. [4%] If we use different input files to do the vector-based power analysis, will the result of rail analysis be different? Why?